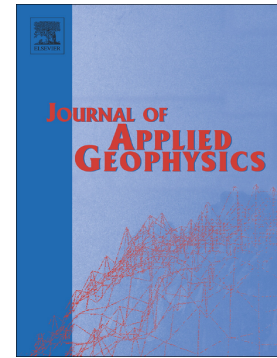


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Time-domain full-waveform inversion of Rayleigh and Love waves in presence of free-surface topography

by

Yudi Pan¹, Lingli Gao² and Thomas Bohlen¹

¹ Geophysical Institute, Karlsruhe Institute of Technology (KIT),
Karlsruhe, Germany.

² School of Earth Sciences, Zhejiang University, Hangzhou, China.

E-mails: yudi.pan@kit.edu;
gaoll1990@126.com;
thomas.bohlen@kit.edu

Corresponding Author:
Yudi Pan (yudipan@yahoo.com or yudi.pan@kit.edu)
Ph. 49 721 608 44606

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ABSTRACT

Correct estimation of near-surface seismic-wave velocity when encountering lateral heterogeneity and free surface topography is one of the challenges to current shallow seismic. We propose to use time-domain full-waveform inversion (FWI) of surface waves, including both Rayleigh and Love waves, to solve this problem. We adopt a 2D time-domain finite-difference method with an improved vacuum formulation (IVF) to simulate shallow-seismic Rayleigh wave in presence of free-surface topography. We modify the IVF for SH-wave equation for the simulation of Love wave in presence of topographic free surface and prove its accuracy by benchmark tests. Checkboard model tests are performed in both case when free-surface topography is included or neglected in FWI. Synthetic model containing a dipping planar free surface and lateral heterogeneity was then tested, in both cases of considering and neglecting free-surface topography. Both checkerboard and synthetic models show that Rayleigh- and Love-wave FWI have similar ability of reconstructing near-surface structures when free-surface topography is considered, while Love-wave FWI could reconstruct near-surface structures better than Rayleigh-wave when free-surface topography is neglected.

Key Words: finite-difference modeling; full-waveform inversion; free surface topography.

INTRODUCTION

The reconstruction of elastic parameters of near-surface materials is of fundamental importance in near-surface geophysical exploration. High-frequency surface-wave techniques have been proved as efficient ways to estimate shallow shear (S)-wave velocity in recent years (Socco et al., 2010). Based on measuring and inverting surface-wave dispersion curves from shallow seismic data, established surface-wave inversion methods assume 1D subsurface structure, and allow the reconstruction of S-wave velocity as a function of depth (Xia et al., 1999; Xia et al., 2012). Different approaches such as laterally constrained inversion (Socco et al., 2009), spatial windowing (Bohlen et al., 2004), and cross-correlation analysis of multi-channel data (Hayashi and Suzuki, 2004) have been proposed to account for lateral heterogeneity. These approaches make surface-wave methods applicable to not only lateral homogeneous but also lateral smooth varying models. However, due to 1D flat layered model assumption involved in the calculation of surface-wave dispersion curves (e.g., Knopoff 1964; Pan et al., 2013), surface-wave methods fail to work when encountering lateral heterogeneity or topographic free surface.

Full-waveform inversion (FWI) was first proposed and formulated in time-domain (e.g., Tarantola, 1984) and later in frequency-domain (Pratt, 1990). Frequency-domain FWI has relatively higher efficiency in calculating the gradient of misfit function, which reduce the computational cost of FWI (Virieux and Operto, 2009). Both time-domain and frequency-domain FWI of shallow seismic wavefield has shown promising prospects of it in reconstructing near-surface elastic parameters (Romdhane et al., 2011; Zeng et al., 2011). Besides synthetic tests, several real-world examples of shallow-seismic FWI have been reported. Tran et al., 2013 showed that it is feasible to detect underground sinkhole via frequency-domain FWI. Köhn et al. (2016) applied a time-domain FWI to an ultrasonic Rayleigh-wave shot gather. Pan et al. (2016)

performed time-domain FWI of Love wave to reconstruct near-surface structures. Groos et al. (2017) proposed a complete multiscale workflow for time-domain FWI of Rayleigh waves, and applied it to a real-world data. Wang and Zhang (2017) proposed a pseudo 2D elastic FWI to obtain near-surface attenuation. Dokter et al. (2017) applied a multiscale time-domain FWI to field SH- (Love-) wave data for characterizing near-surface heterogeneity. Most of the studies regarding time-domain FWI are conducted with a planar free surface. However, near-surface media can be highly complex, where topographic free surface is commonly seen. Nuber et al. (2016) demonstrated through synthetic tests that even minor topographic variation of free surface would have a significant effect on FWI.

The forward modeling of surface waves is of fundamental importance to high-frequency surface-wave techniques and FWI. The finite-difference method (FDM) has been widely used in the numerical simulation of seismic waves for its high efficiency and accuracy (Bohlen, 2002). Free surface traction-free condition (Aki and Richards, 2002) is critical to the accuracy of finite-difference (FD) modeling. Rayleigh wave can be simulated by solving the 2D P-SV elastic-wave equation (Virieux, 1986), while Love wave are simulated by solving the 2D SH-wave elastic-wave equation (Virieux, 1984) with a suitable free surface condition.

FD modeling of Rayleigh wave with planar or topographic free surface has been well studied. Levander (1988) proposed a stress-image technique to satisfy the free-surface condition. Robertsson (1996) and Graves (1996) extended the classical stress-image technique to an arbitrary free-surface topography and 3D planar free-surface situations, respectively. The accuracy of the stress-image technique has been tested and improved by adjusted FD approximations (Kristek et al., 2002, Moczo et al., 2004). Xu et al. (2007) proposed an acoustic/elastic interface approach to simulate planar free surface, which is further developed for

topographic free surface for 2D and 3D situations (Wang et al., 2015). Gao et al. (2015) proposed an immersed-boundary method to simulate topographic free surface, however, with relatively lower stability. The vacuum formulation, which is first proposed by Zahradník et al. (1993), and improved by Zeng et al. (2012) provided an easy and efficient way for the FD modeling of Rayleigh wave in presence of free surface topography.

Most researches on FD modeling of surface waves focus on Rayleigh waves, only a few contributions have been performed on Love wave. Kelly (1983) performed a numerical study on the propagation of Love wave with a standard-grid FDM. Virieux (1984) proposed a velocity-stress staggered-grid FDM for the forward modeling of SH waves. Narayan and Kumar (2008) used a variable grid-size FDM for the forward modeling of SH waves. Kumar and Narayan (2008) implemented the absorbing boundary condition to the variable grid-size SH-wave FDM. Luo et al. (2010) used FDM to model and study the characteristics of shallow Love wave. For the increasing demand of studying and using Love wave in shallow seismic, accurate forward modeling algorithm for Love wave in presence of surface topography is required.

We propose to resolve near-surface S-wave velocity structures by performing time-domain FWI of shallow seismic surface waves. A 2D finite-difference method with an improved vacuum formulation (IVF; Zeng et al., 2012) is adopted as the forward solver for Rayleigh-wave simulation, and modified for the simulation of Love wave. We perform two benchmark tests comparing to the Spectral-Element Method (SEM) (Komatitsch and Tromp, 1999) to prove the accuracy of the modified method. A conjugate-gradient algorithm is used for optimization in the FWI. Checkerboard models with planar and topographic free surfaces are used to test the influence of the topographic free surface on surface-wave FWI. A synthetic example with dipping planar free surface is then tested in both cases of including and neglecting free-surface

topography. Rayleigh- and Love-wave FWI results are compared in both checkerboard model and synthetic model tests, showing some advantages of using Love-wave FWI for reconstructing near-surface structures.

FORWARD MODELING

Efficient and accurate simulation of shallow seismic wavefields is essential to FWI. Rayleigh wave are formed by the interfering of P and SV waves governing by a 2D P-SV elastic-wave equation (Virieux, 1986), while Love wave are formed by the interfering of SH waves governing by a 2D SH elastic-wave equation (Virieux, 1984).

2D P-SV elastic-wave equation can be written as:

$$\frac{\partial v_x}{\partial t} = b \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xz}}{\partial z} \right) \quad (1)$$

$$\frac{\partial v_z}{\partial t} = b \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{zz}}{\partial z} \right) \quad (2)$$

$$\frac{\partial \tau_{xx}}{\partial t} = (\lambda + 2\mu) \frac{\partial v_x}{\partial x} + \lambda \frac{\partial v_z}{\partial z} \quad (3)$$

$$\frac{\partial \tau_{zz}}{\partial t} = (\lambda + 2\mu) \frac{\partial v_z}{\partial z} + \lambda \frac{\partial v_x}{\partial x} \quad (4)$$

$$\frac{\partial \tau_{xz}}{\partial t} = \mu \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \quad (5)$$

2D SH elastic-wave equation can be written as:

$$\frac{\partial v_y}{\partial t} = b \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{zy}}{\partial z} \right) \quad (6)$$

$$\frac{\partial \tau_{xy}}{\partial t} = \mu \frac{\partial v_y}{\partial x} \quad (7)$$

$$\frac{\partial \tau_{zy}}{\partial t} = \mu \frac{\partial v_y}{\partial z} \quad (8)$$

where v_x , v_y and v_z represent the particle velocity, b buoyancy (reciprocal of density ρ), $\mu(x,z)$ and $\lambda(x,z)$ the Lamé coefficients, τ_{xx} , τ_{xz} , τ_{zz} , τ_{xy} and τ_{zy} are the shear stresses.

For the forward modeling of synthetic waveforms, we use a second-order time derivative and fourth-order space derivative staggered-grid FDM (Virieux, 1984 and 1986). The free-surface boundary condition is critical for the correct modeling of surface waves, especially when encountering topographic free surface. In FD modeling, the shape of free-surface topography can be approximated by staircase shape (Hayashi et al., 2001) with small vertical and horizontal planar segments. Zeng et al. (2012) proposed an IVF to simulate Rayleigh wave (P-SV waves) in presence of free-surface topography with high efficiency. We adopted this method for the forward modeling of Rayleigh wave and modify it in the case of Love wave (SH waves).

In the IVF, a two-step pre-processing scheme is used for the simulation of free-surface topography. The first step is the parameter averaging where the effective shear modulus and effective buoyancy are obtained as follows

For P-SV waves:

$$b_x = \begin{cases} 0, & \text{if } \rho_{i,k} = 0 \text{ and } \rho_{i+1,k} = 0; \\ \frac{2}{\rho_{i,k} + \rho_{i+1,k}}, & \text{otherwise;} \end{cases} \quad (9)$$

$$b_z = \begin{cases} 0, & \text{if } \rho_{i,k} = 0 \text{ and } \rho_{i,k+1} = 0; \\ \frac{2}{\rho_{i,k} + \rho_{i,k+1}}, & \text{otherwise;} \end{cases} \quad (10)$$

$$\mu_{kz} = \begin{cases} \left[\frac{1}{4} \left(\frac{1}{\mu_{i,k}} + \frac{1}{\mu_{i+1,k}} + \frac{1}{\mu_{i,k+1}} + \frac{1}{\mu_{i+1,k+1}} \right) \right]^{-1}, & \text{if } \mu_{i,k} \mu_{i+1,k} \mu_{i,k+1} \mu_{i+1,k+1} \neq 0; \\ 0, & \text{otherwise;} \end{cases} \quad (11)$$

For SH waves:

$$b_y = \begin{cases} 0, & \text{if } \rho_{i,k} = 0; \\ \frac{1}{\rho_{i,k}}, & \text{otherwise.} \end{cases} \quad (12)$$

$$\mu_x = \begin{cases} \left[\frac{1}{2} \left(\frac{1}{\mu_{i,k}} + \frac{1}{\mu_{i+1,k}} \right) \right]^{-1}, & \text{if } \mu_{i,k} \mu_{i+1,k} \neq 0; \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

$$\mu_z = \begin{cases} \left[\frac{1}{2} \left(\frac{1}{\mu_{i,k}} + \frac{1}{\mu_{i,k+1}} \right) \right]^{-1}, & \text{if } \mu_{i,k} \mu_{i,k+1} \neq 0; \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

where b_x , b_y , and b_z represent the effective buoyancy, and μ_{xz} , μ_x and μ_z are the effective shear moduli, respectively.

The second step in the pre-processing scheme of IVF is setting a half-grid-thick fictitious layer above the original topographic surface (area in shadow in Figure 1). The surface on the top of the fictitious layer is regarded as the free surface during the FD calculation (bold solid lines in Figure 1). Material above (or inside in the case of internal discontinuities) the free surface (bold line in Figure 1) is set as vacuum ($\rho = 0$, $\mu = 0$). The fictitious layer is used for analysis purposes, and its physical properties are calculated using the adjacent grids with the average scheme presented above (equations 9-14). As a result, the effective shear modulus μ_{xz} is always zero on the free-surface boundary line of the fictitious layer in the case of P-SV waves, and the effective shear moduli μ_x and μ_z is always zero on vertical (points A, D, E, H in Figure 1) and horizontal (points B, C, F, G in Figure 1) free-surface segments, respectively, in the case of SH waves.

After these two steps of pre-processing, the free-surface condition would be automatically fulfilled during the FD calculation. In the case of P-SV waves, only τ_{xz} on the free surface needs to fulfill the traction-free boundary condition (e.g., inverted triangle between points A and B in Figure 1). Because all the effective shear modulus μ_{xz} become zero on the free surface boundary line after parameters averaging (equation 11), τ_{xz} on the free surface is automatically zero during the calculation according to equation 5. Similarly, in the case of SH waves, all effective moduli μ_x are zeros along the vertical free surface, where only τ_{xy} are placed (e.g., rectangles at points A,

D, E and H in Figure 1). Therefore, all the stress components τ_{xy} along the free surface become automatically zero during the FD calculation (equation 7). Meanwhile, all the zero-value effective moduli μ_z along the horizontal free surface make all the stress components τ_{zy} (e.g., circle at points B, C, F and G in Figure 1) zero during the FD calculation (equation 8). The averaged buoyancy on the free surface is twice the buoyancy at the adjacent grid in the solid earth in the case of P-SV waves, which is consistent with the elastic-parameter modification scheme for planar free surface (Mittet, 2002; Xu et al., 2007). This makes IVF a universal way for the simulation of free-surface boundary condition.

BENCHMARK

Analytical solutions are usually for the benchmark of numerical modeling results (e.g., Bohlen and Saenger, 2006). However, analytical or semi-analytical solution of SH waves only exist with horizontal free surface or quarter-plane problem (Wait 1959; Wang et al., 2012), which makes it unable to use analytical solution for the benchmark in our case. SEM naturally satisfies the free-surface condition by approximating the topographic surface by tetrahedral or hexahedral meshes, which provides an accurate way for the numerical simulation of surface wave regardless of free-surface topography (Komatitsch and Tromp, 1999). In this paper, we perform the benchmark tests by comparing the synthetic waveforms calculated by the proposed method to those calculated by SEM. The accuracy of IVF for the simulating of Rayleigh wave in presence of free-surface topography has been proven by Zeng et al. (2012). Here we only perform benchmark to prove the accuracy of using IVF to simulate Love wave.

The first benchmark model is a two-layer model with a topographic free surface and a non-planar interface between two layers (Figure 2). The S-wave velocities of the two layers are 200

m/s and 400 m/s, respectively. The densities of the two layers are 1950 kg/m^3 and 2050 kg/m^3 , respectively. Both the shapes of the free surface and that of the layer interface are irregular, while the thickness of the first layer ranges from 5.5 m to 10 m.

We use a fine grid of $0.1 \text{ m} \times 0.1 \text{ m}$ to discretize the model in order to reduce the influences caused by the staircase shape (Hayashi et al., 2001). The time step is 0.02 ms to ensure the stability of FD modeling. Perfectly Matched Layers (PML) are applied to the left, right, and bottom of the model with a thickness of 10 m (Collino and Tsogka, 2001). Ten traces of transverse-component receivers are placed along the free surface. Both the nearest source-receiver offset and trace interval are 5 m along horizontal direction (x axis in Figure 3). A 20 Hz Ricker wavelet is used as the source time signal.

Waveforms of the 5th and 10th traces, which are placed 25 m and 50 m away from the source are compared (Figure 3a and b). The synthetic waveforms calculated by FDM and SEM are almost identical (dashed and solid lines in Figure 3), which proves the high accuracy of IVF for Love wave modeling. For comparison, we also calculate the synthetic waveforms calculated by FDM but with the conventional vacuum formulation (CVF) in which shear moduli are not averaged and no fictitious layer is added above the original free surface (dotted line in Figure 3). The synthetic waveforms calculated by CVF cannot fit the other two, which shows the improvement of accuracy in Love-wave modeling by using IVF compared to CVF.

The IVF can simulate not only topographic free surface, but also internal discontinuities. The second benchmark model is a two-layer model with two rectangle tunnels inside the first layer (Figure 4). The sizes of the two tunnels are $3 \text{ m} \times 3 \text{ m}$ and $4 \text{ m} \times 4 \text{ m}$, respectively. The S-wave velocities of the two layers are 200 m/s and 400 m/s, respectively. The density is 2000

kg/m³ for both layers. The spatial and time steps are the same with the previous benchmark test. The source temporal signal is a 30 Hz Ricker wavelet.

Forty traces of receivers are placed along the free surface with a nearest offset and a trace interval of 1 m. Diffracted Love wave (or SH wave) can be clearly identified in the synthetic shot gather between 120 ms and 280 ms (Figure 5a). Synthetic waveforms calculated by FDM and SEM show high similarity with each other, indicating high accuracy of the proposed method. Strong diffracted Love wave (or SH wave) exist between 150 ms and 220 ms in the synthetic waveform, which are caused by the two tunnels. Fluctuation in the waveform after 220 ms is mainly caused by multiple reflections of diffracted Love wave (or SH wave) between the free surface and the tunnels. The simulated waveforms of the 16th trace, which is placed between the two tunnels (inverted triangle in Figure 4), are compared (Figure 5b). Waveforms simulated by FDM and SEM are almost identical. The waveform simulated by FDM has a slight phase advance in the refracted surface wave comparing to that modeled by SEM (waveform after 150 ms in Figure 5b). This is caused by the half-grid-thick fictitious layer that added outside the void in FDM, which slightly enlarges the sizes of voids in the model. The comparison of numerical waveforms with SEM proves that Love-wave wavefields can be correctly simulated by the proposed method.

INVERSE PROBLEM

FWI is getting increasingly popular in shallow seismic due to its ability in reconstructing a high-resolution model of the material parameter in the subsurface. By minimizing the misfit between observed and synthetic waveform, the misfit function in FWI is usually defined via an L2-norm as:

$$E(\mathbf{m}) = \frac{1}{2} \|\mathbf{u}_{syn} - \mathbf{u}_{obs}\|^2 \quad (15)$$

where $\mathbf{u}_{syn}$ and $\mathbf{u}_{obs}$ represent the synthetic and observed wavefield. $\mathbf{m}$ represents the model parameters. Synthetic and observed waveforms are normalized at each trace.

The model parameter is updated iteratively to find a model $\mathbf{m}$ which can explain the observed data well:

$$\mathbf{m}_{k+1} = \mathbf{m}_k + \alpha_k \Delta \mathbf{m}_k \quad (16)$$

where $\mathbf{m}_{k+1}$ and $\mathbf{m}_k$ represent model at the $(k+1)$ th and k th iteration, α_k represents the step length at the k th iteration, which is determined by a combination of line search and parabolic fitting, $\Delta \mathbf{m}_k$ represents the model update, which can be obtained via a conjugate gradient (CG) algorithm:

$$\Delta \mathbf{m}_k = -H_a^{-1} \Delta \mathbf{g}_k + \beta_k \Delta \mathbf{m}_{k-1} \quad (17)$$

where $\Delta \mathbf{g}_k$ denotes the gradient of objective function E with respect to the model $\mathbf{m}$. H_a^{-1} represents the inverse of approximated Hessian matrix by using its diagonal term (Plessix and Mulder, 2004). β_k is determined by:

$$\beta_k^{HS} = \max \left\{ \frac{\Delta \mathbf{g}_k^T (\Delta \mathbf{g}_k - \Delta \mathbf{g}_{k-1})}{\Delta \mathbf{m}_{k-1}^T (\Delta \mathbf{g}_k - \Delta \mathbf{g}_{k-1})}, 0 \right\} \quad (18)$$

We adopt an HS (Hestenes and Stiefel, 1952) version of CG algorithm because it is one of the most efficient versions of CG in the time-domain FWI (Liu et al., 2017). Due to a great number of model parameters, the direct calculating of gradient at every single model parameter is computational expensive. An effective way to calculate it is to use adjoint state algorithm (Tarantola 1984; Mora, 1987). A detailed review on the adjoint state algorithm can be found in Plessix (2006). By using the adjoint state algorithm, the gradient of misfit function can be obtained by the zero-lag cross correlation of forward propagating wavefield $\mathbf{u}$ that is generated by the source time function at the source, and its adjoint wavefield $\mathbf{u}^*$ which is calculated by the

back-propagation of residual wavefield at the receivers' positions. A two-grid and a five-grid precondition taper are applied to the vicinity of free surface and source positions to mitigate free-surface and source artefacts (Gross et al., 2017).

CHECKERBOARD MODEL TEST

We used checkerboard models to identify the influence of free surface topography. The background model is a two-layer model with an S-wave velocity of 200 m/s and 400 m/s, and a P-wave velocity of 500 m/s and 1000 m/s, respectively. The density is 2000 kg/m³ for the whole model. The thickness of the first layer is 10 m from the average elevation of free surface. The checkerboard models are formed by blocks of 2 m x 2 m, and the S-wave velocity of each block is 160 or 240 m/s depending on its position, i.e. it differs by 20% relative to the background model (Figure 6a), and the P-wave velocity and density models are kept unchanged. Forty geophones are placed along the free surface with an interval of 1 m (along the free surface shown in Figure 6a). Five sources (vertical sources for Rayleigh wave and horizontal sources for Love wave) are simulated with an interval of 11 m, respectively. The leftmost source is 3 m away on the left side of the first trace and the rightmost source is 2 m away on the right side of the last trace.

Fined grid (0.2 m) and time step (0.05 ms) are used to ensure the accurate simulation of surface waves with topographic free surface. PML are appended at the outside of the physical model to avoid artificial reflections. A 40 Hz Ricker wavelet is used as source, and is assumed as known during the inversion. Minimal wavelength of surface wave is around 2 m, which makes the size of checkerboard block (2 m x 2 m) within the resolution of FWI (half of the wavelength). The P-wave velocity and density are set as the true model (the two-layer model) and not updated

during the inversion. So only S-wave velocity is updated during the inversion for both Rayleigh and Love waves. Both vertical and inline-horizontal components are used in Rayleigh-wave FWI, while crossline-horizontal component is used in Love-wave FWI.

The true model contains a sinkhole of 5 m wide and 1 m deep on the planar free surface (Figure 6a). The two-layer background model is used as the initial model. Both synthetic tests including and neglecting free-surface topography are performed. The misfit values converge after 50 iterations, while Love-wave FWI converges at relatively smaller misfit values comparing to Rayleigh wave (Figure 6b).

Checkerboard blocks are well or poorly reconstructed when the free-surface topography is included or neglected (first and second rows in Figure 6a), respectively. When the topography is included, Love- and Rayleigh-wave FWI results show similar accuracy, which indicates similar robustness of Love- and Rayleigh-wave FWI when the free-surface topography is exactly known. When the topography is neglected, checkerboard blocks are better reconstructed by using Love waves compared to Rayleigh waves. The reconstructed checkerboard blocks are clearer in the subsurface areas below the free surface where is planar compared to the topographic part. Both Love- and Rayleigh-wave FWI results contain strong artefacts at the two edges of the “missing” sinkhole ($X = 15$ and 20 m at the free surface), which try to fit the scattered waves that caused by the free-surface topography in the observed data. This synthetic example shows that ignoring free-surface topography distorts the FWI result, and Love-wave FWI result is less distorted compared to Rayleigh-wave when free-surface topography is not exactly known. We may improve the Rayleigh-wave FWI result by rotating the observed data acquired along the topographic free surface to compensate for the topography. However, the rotation requires

information about topography, while it is better to include the topography directly in FWI if the topography is known.

DIPPING PLANAR SURFACE MODEL

Another synthetic example is performed by using a model containing a dipping planar free surface with a slope of 10 degrees (Figure 7). The model consists of three layers with two non-planar interfaces in between, a circular low-velocity body and a triangular high-velocity body with a S-wave velocity of 150 m/s and 400 m/s, respectively. Homogeneous half space models are used for P-wave velocity and density, with a value of 600 m/s and 2000 kg/m³, respectively. Forty receivers are placed along the free surface (dashed line on the free surface in Figure 7), and five sources (vertical and horizontal sources for Rayleigh and Love waves, respectively) are generated (stars on the free surface in Figure 7), respectively. A 0.1 m fine grid and a 0.05 ms time step are used for FDM. A linear gradient velocity model is set as the initial model (Figure 8). The P-wave velocity, density, and source wavelet are set as the true model (homogeneous half-space models and a 25 Hz Ricker wavelet with 50 ms delay) and are not updated during the inversion, in order to exclude their influences in reconstructing S-wave velocity model.

Distinguished difference can be identified between the synthetic waveforms due to initial model and the observed waveform, while the observed waveform is almost perfectly fitted by the synthetic waveform of inversion result after 100 iterations (Figure 9a). Though a same initial model is adopted for Love and Rayleigh waves, the misfit between observed and synthetic waveforms of initial model is greater in Rayleigh wave comparing to Love wave. Love-wave waveform is relatively simpler than that of Rayleigh wave, which reduces the possibility of cycle skipping in Love-wave FWI. Besides, the refracted wave caused by the high-velocity body is

more obvious in the Love-wave waveform (between 160 ms to 280 ms in the first 20 traces in Figure 9a), indicating a higher sensitivity of the Love-wave FWI to the high-velocity body. In both results of Love- and Rayleigh-wave FWI, the S-wave velocity models are well reconstructed (Figure 9b) in which all the main structures including low- and high-velocity bodies and non-planar layer interfaces (dashed lines in Figure 9b) are clearly delineated. Results of Love- and Rayleigh-wave FWI show similar accuracy, indicating that they work equally well when encountering dipping planar free surface while the topographic free surface is correctly included during the inversion.

The second test is performed on the same model by neglecting the free surface topography. In other word, a horizontal planar free surface is used for the inversion. A similar linear gradient velocity model is used as the initial model, however, rotated 10 degrees anticlockwise (Figure 8).

Though the dipping of free surface is neglected during the inversion, the main structures of the model, including the existences of high- and low-velocity bodies and the shapes of layer interfaces, are reconstructed quite well especially in the case of Love wave (Figure 10a). In the Rayleigh-wave FWI result (Figure 10b), the velocities for the high- and low-velocity bodies are underestimated, while the uplift of bedrock is not clearly delineated due to the neglecting of free-surface topography. In the Love-wave FWI result, however, all the main structures are well reconstructed with correct positions and velocities. This is because 2D SH wave elastic-wave equations are independent to the 2D Cartesian coordinate, which makes Love wave invariant to rotations of the free-surface topography. It indicates that it is promising to use Love wave when encountering dipping planar free surface for its directional independency. An inversion result with higher accuracy can be obtained if we rotate both model and observed data in the Rayleigh-wave case (Figure 10c). However, the rotation of observed data requires an accurate knowledge

about the dipping angle of free surface, while this rotation is not needed in the case of Love waves. Some artefacts can be identified around the free surface in the Love-wave FWI result. They are mainly caused by the numerical artefacts in the “observed” data in which dipping planar free surface is not perfect planar caused by the influence of staircase FD grid. These artefacts can be mitigated or removed by applying a stronger precondition taper to the vicinity of free surface.

Nuber et al. (2016) highlighted the particularly importance of including free surface topography in shallow-seismic FWI. Due to the invariance of Love wave to the rotations of the free-surface topography, however, our synthetic results show that in case of dipping planar free surface, acceptable result can be obtained even free surface topography is neglected during the Love-wave FWI. This result, on one hand, serves as a complement to the conclusions of Nuber et al. (2016), and on the other hand, highlights the advantages of utilizing Love wave in shallow seismic (Dokter et al., 2017) especially when encountering free-surface topography.

CONCLUSIONS

We performed a time-domain FWI of shallow-seismic surface waves, including both Rayleigh and Love waves, when encountering topographic free surfaces. An improved vacuum formulation (IVF) was adopted and developed for the simulation of surface waves. Benchmark proved the relative high accuracy of the IVF for simulating Love wave in presence of free-surface topography. Checkerboard model tests showed that the existence of an irregular free surface would not affect (or only slightly reduce) the resolution of inversion result if it can be correctly included during the inversion. A synthetic model containing a dipping planar free surface was tested, which proved the validity of performing time-domain FWI to delineate near-surface S-wave velocity

structures when encountering both lateral heterogeneity and surface topography. Rayleigh and Love wave show similar resolution (half of the wavelength) of reconstructing underground S-wave velocity structures in checkerboard and synthetic model tests. Due to the simpler waveform and less influences by the topographic free surface, it is more promising to use Love wave in shallow-seismic FWI. Further studies on application to real-world data set with strong free-surface topography are required.

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FIGURE CAPTIONS

Figure 1. Grid distribution for the improved vacuum formulation. The area in grey and shadow represent earth medium and a fictitious half-grid-thick layer, respectively. The line in bold represents the free surface for finite-difference calculation. Solid line represent grid that exists in both P-SV and SH wave equations. Dashed line represent grid that only exists in P-SV wave equation.

Figure 2. First benchmark model with a topographic free surface and a non-planar interface. Star and inverted triangles represent transverse source and receivers, respectively.

Figure 3. Comparison between synthetic waveforms acquired at a) 5th and b) 10th trace in Figure 2. Solid line represents synthetic waveforms calculated by spectral element method. Dashed and dotted lines represent synthetic waveforms calculated by finite difference method with improved and conventional vacuum formulations, respectively.

Figure 4. Second benchmark model with two tunnels. Forty receivers are placed along the free surface with a nearest offset and a trace interval of 1 m, respectively. The inverted triangle represents the 16th trace, which is placed between the two tunnels.

Figure 5. (a) Synthetic shot gathers calculated by spectral element method and finite difference method. Waveforms are displayed every second trace. (b) Comparison on the 16th trace which is placed between the two tunnels (see Figure 5). The waveforms are normalized in each trace.

Figure 6. (a) Results of Love- (left) and Rayleigh-wave (right) FWI. Two columns represent FWI results when including and neglecting free-surface topography, respectively. (b) Change of misfit values with iterations.

Figure 7. True S-wave velocity model for the dipping planar free surface model. Dashed line and stars on the free surface represent receivers and sources, respectively.

Figure 8. Initial linear gradient S-wave velocity model for the synthetic test.

Figure 9. (a) Observed waveform and synthetic waveforms of initial model and inversion result. Waveforms are displayed every second trace. (b) FWI results of Love (left) and Rayleigh waves (right) when considering free-surface topography. Dashed lines represent the location of main structures.

Figure 10. FWI results of a) raw observed Love waves, b) raw observed Rayleigh waves, and c) rotated observed Rayleigh waves by neglecting free-surface topography. Dashed lines represent the location of main structures.

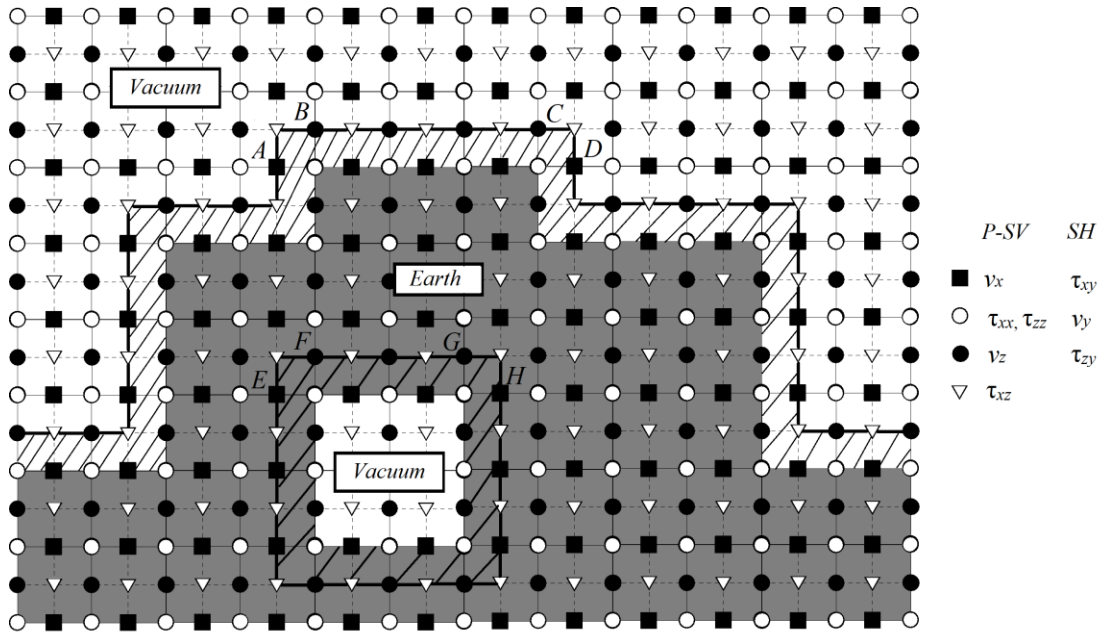


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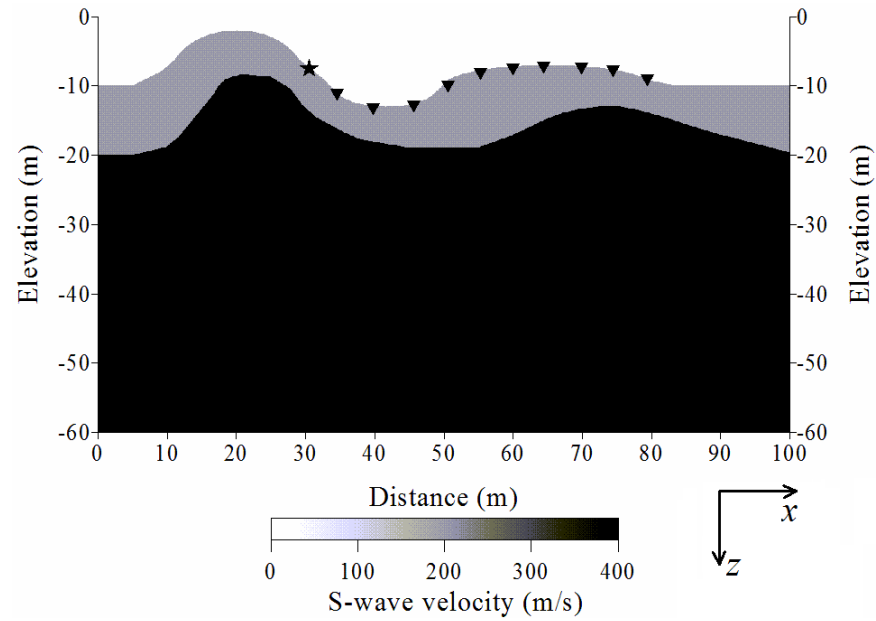


Figure 2. First benchmark model with a topographic free surface and a non-planar interface. Star and inverted triangles represent transverse source and receivers, respectively.

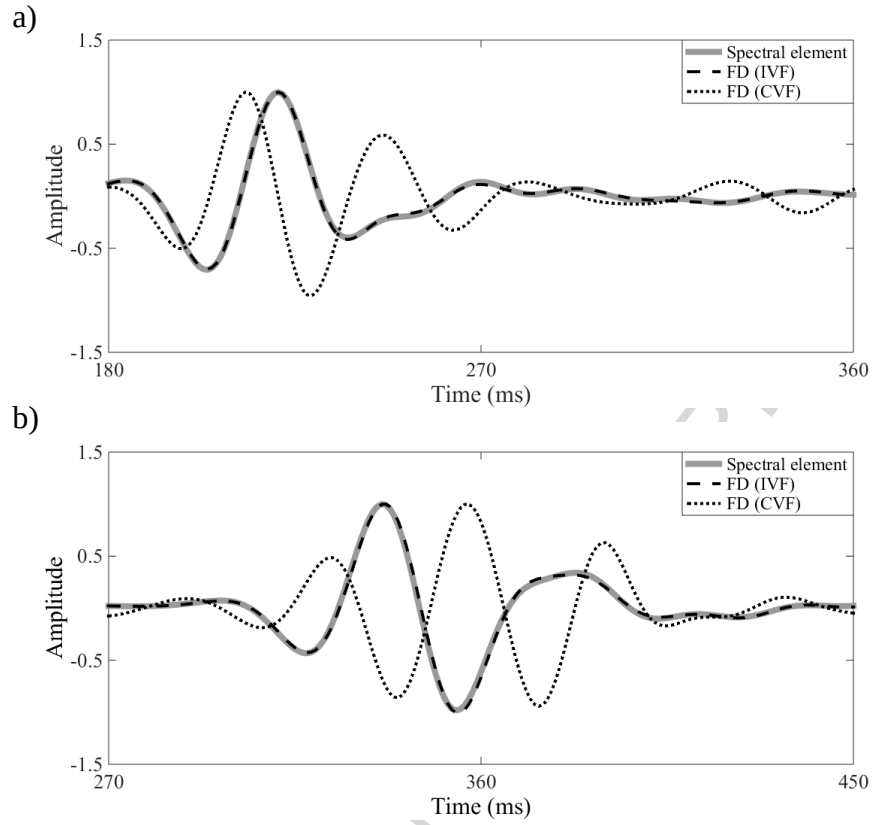


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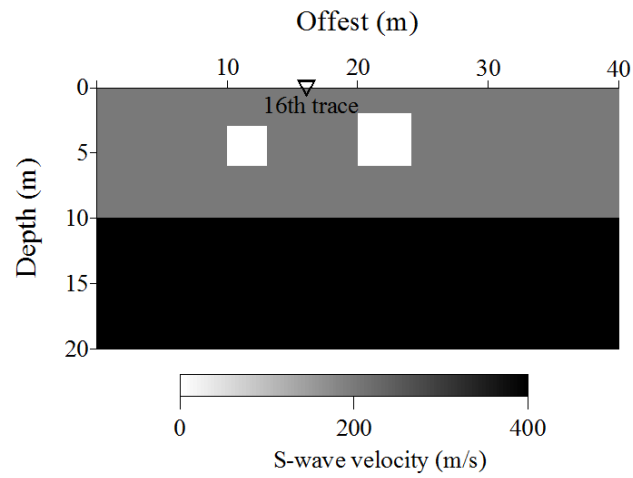


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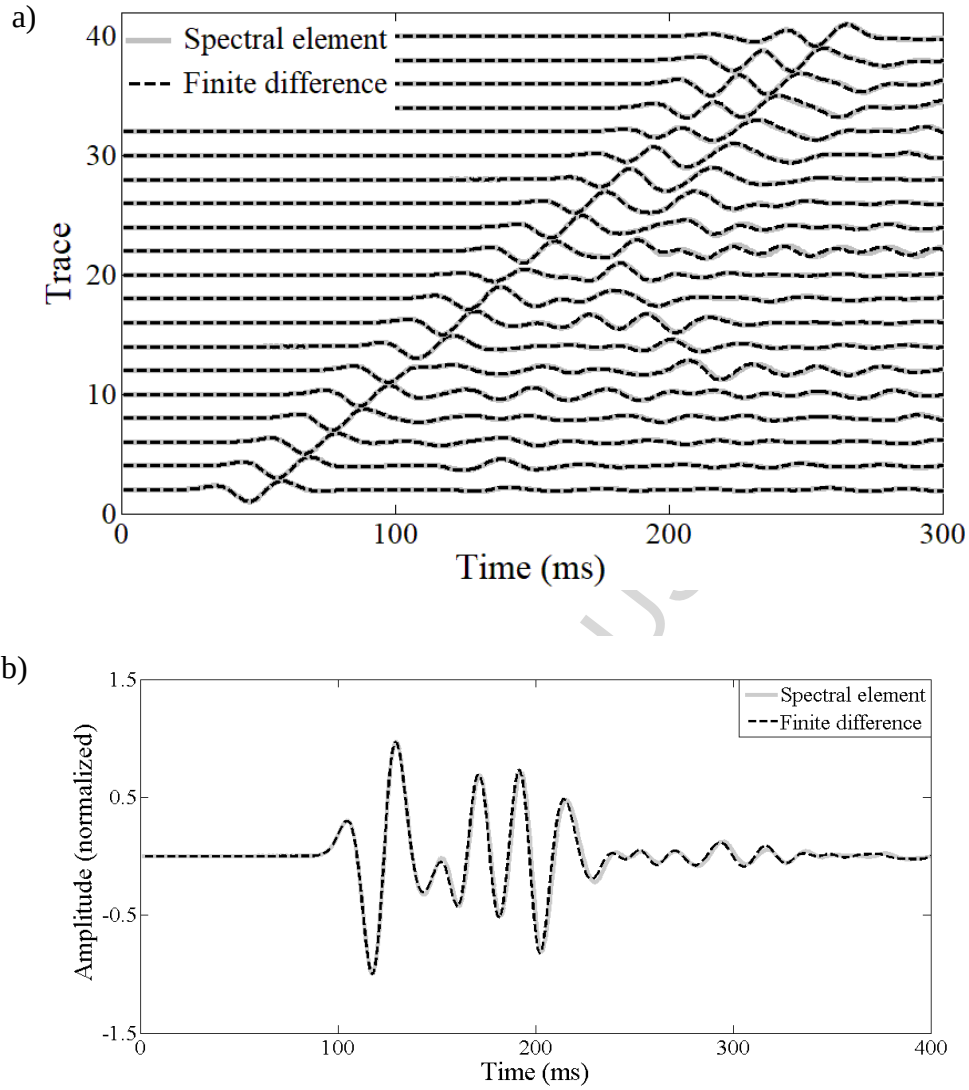


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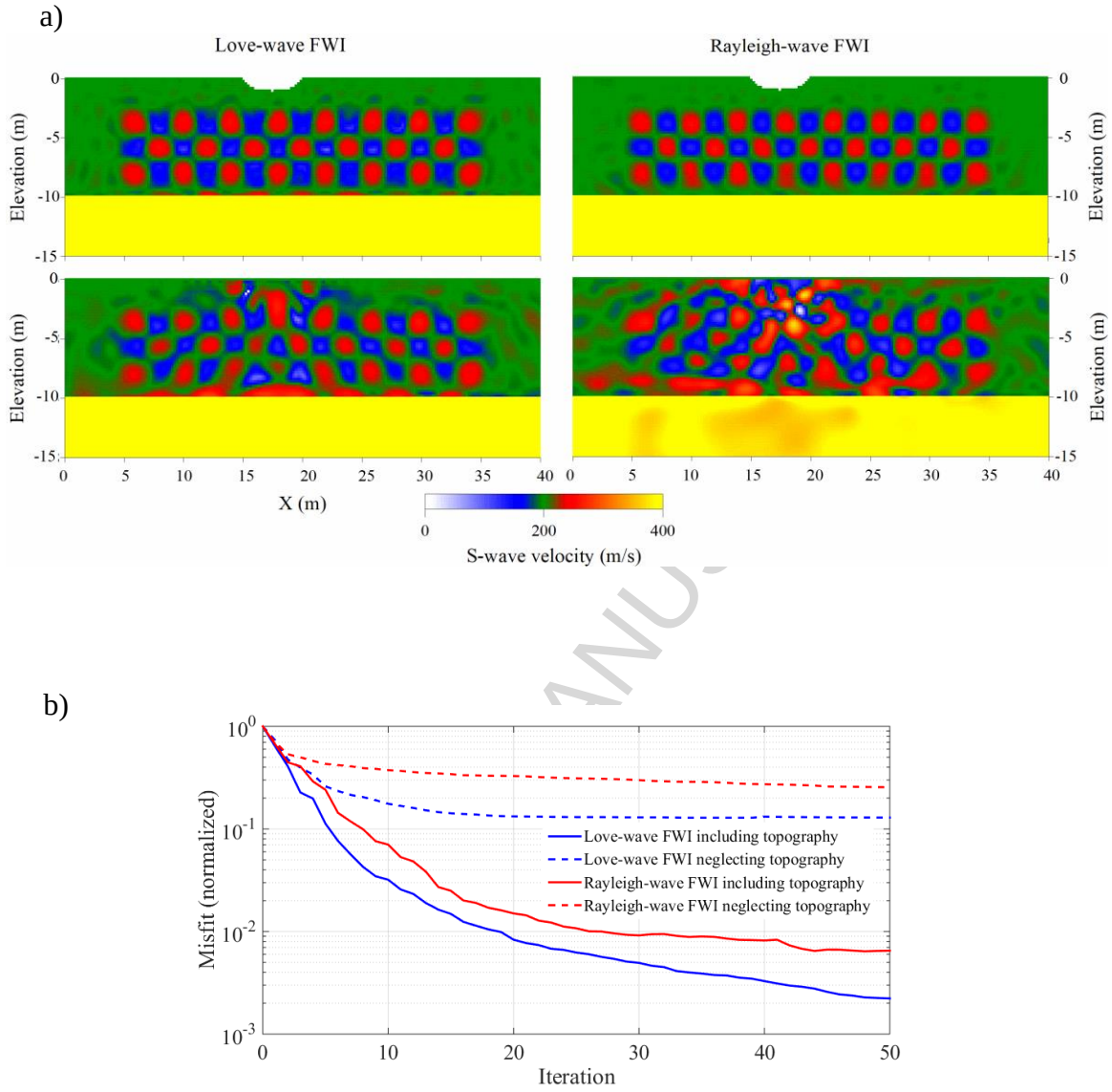


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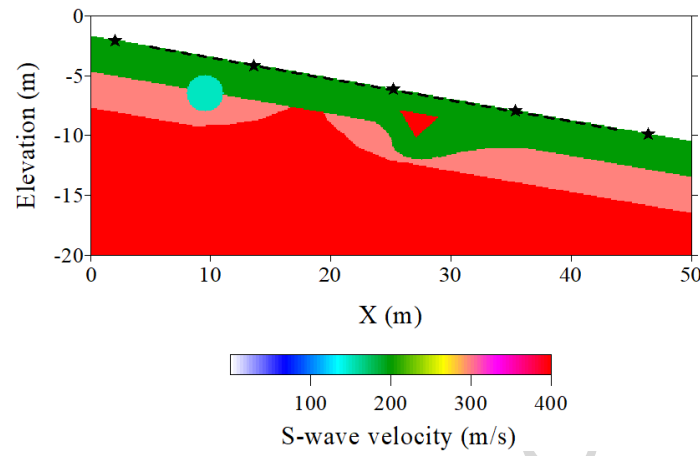


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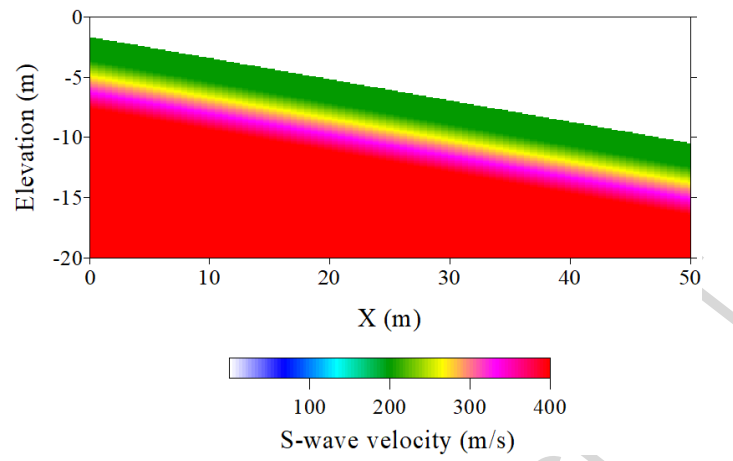


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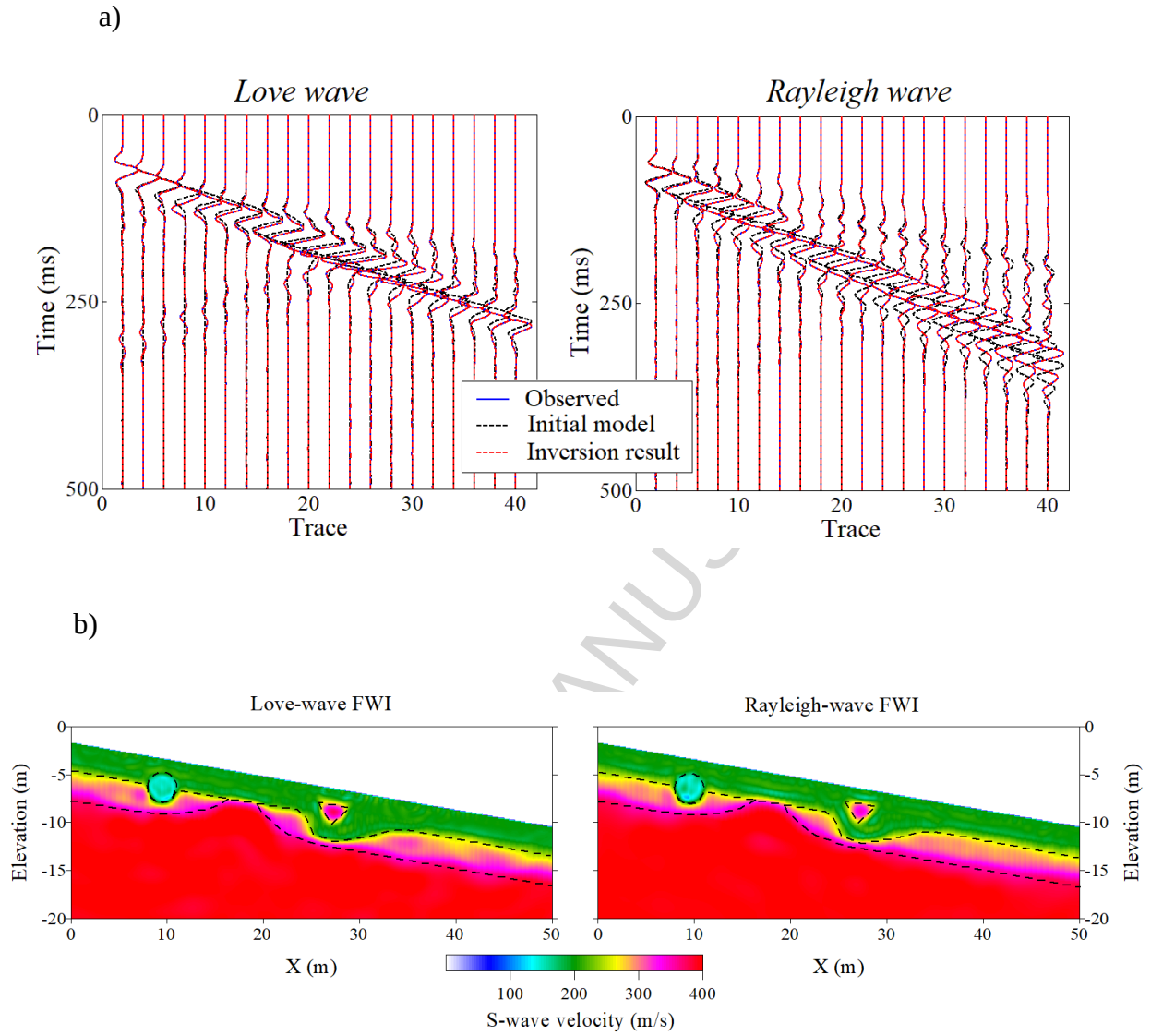


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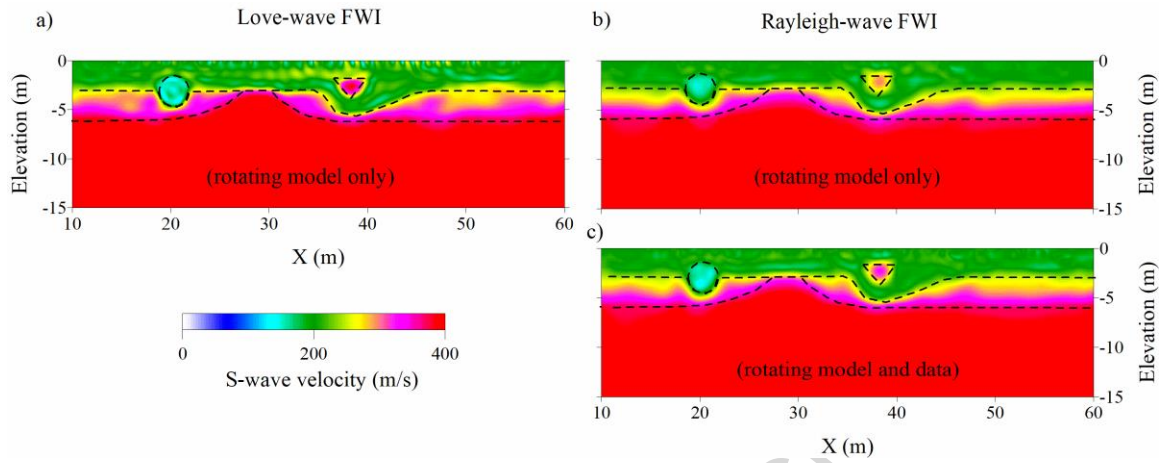


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Highlights:

- An improved vacuum formulation for simulating free-surface topography.
- Influence of free-surface topography on the resolution of surface-wave FWI.
- Comparison between Rayleigh- and Love-wave FWI.

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